SÉBASTIEN BOULNOIS, Ph.D.

Human Factors & Human-Systems Integration Engineer

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PROFESSIONAL SUMMARY

Human Factors and Human-Systems Integration researcher with a Ph.D. in Human-Centered Design (aviation) and 10+ years of experience studying how operators make decisions in high-consequence systems. Designed and ran human-in-the-loop studies in B737 and A320 cockpit simulators with 20+ expert pilots, assessing workload (NASA-TLX), situation awareness (SART), and decision-making. Conducted field research in three nuclear power plants and five years of applied research for Fortune 500 clients. Skilled in cognitive task analysis, scenario-based design, prototyping, and human-machine interface evaluation. Translates operator goals, workflows, challenges, and workarounds into design requirements and recommendations that engineers and program leaders can act on.

EXPERIENCE

Chief Operating Officer (Part-Time)
Bee Humans, Baer Hive, Inc. (early-stage startup) - Palmdale, CA - 2026 - Present

- Developing vision and strategy for Human-Centered Design, Human-Systems Integration and Human Factors operations in Apiculture innovation.
- Developing a deep understanding of commercial pollinators' field operations and acute challenges.
- Informing innovative products that support beekeepers' health and performance.

Career Break - Family Caregiving
Palmdale, CA - 2024–2026

Senior Researcher
UST Global - San Diego, CA - 2024

- Developed and operationalized rapid research methodologies to support evidence-based decision-making within agile product teams.
- Led participant recruitment and qualitative research with high-profile users to inform strategic product decisions for one of the world's largest electronic marketplaces.

Researcher, promoted to Senior Researcher (2022)
Round Feather - San Diego, CA - 2019–2024

- Led and contributed to 17 generative research initiatives for Fortune 500 clients, uncovering people's goals, the tools available to them, and the challenges, workarounds, and emotions that shaped their decisions across end-to-end journeys, to inform product strategy and development roadmaps.
- Designed and institutionalized a continuous user-testing program for Meta's employee

financial well-being application (Paven), conducting 100+ evaluative sessions that informed iterative improvements in usability, workflow design, and human-system interaction.

- Led end-to-end research, prototyping, and usability testing for a self-service platform supporting individuals through unexpected pregnancies, contributing to scaled engagement reaching 35K+ users, with 10K+ using its interactive features.
- Applied in-depth interviews, qualitative journal studies, task and activity analysis, and journey and workflow modeling to pinpoint decision breakdowns, then proposed new or alternative products, processes, and services that made complex decisions easier for end users.

Researcher
Daito Design - Melbourne, FL - 2019

- Conducted field research across three nuclear power plants to identify human-system barriers affecting adoption of a Digital Plant Viewer in a highly regulated environment.
- Employed contextual observations, structured interviews, task analysis, and usability testing to elicit user requirements and characterize operational workflows.
- Delivered research-driven concepts and early-stage design recommendations that informed product strategy and development priorities.

Ph.D. Researcher - Human-Centered Design (Aviation)
Florida Institute of Technology - 2015–2018

- Conducted doctoral research on human-systems interaction in safety-critical cockpit environments, including requirements elicitation, prototyping, evaluation, and refinement of a 3D weather awareness tool compared against traditional onboard weather radar.
- Designed and executed human-in-the-loop studies using B737 and A320 simulators with 20+ expert pilots, collecting and analyzing workload (NASA-TLX), situation awareness (SART), usability (SUS), and decision-making data to evaluate pilot interaction with onboard weather management systems.
- Collaborated with aviation industry stakeholders and research institutions to align experimental methods with operational realities and safety requirements.
- Applied semi-structured interviews, grounded theory, cognitive task analysis, participatory design, and scenario-based design to derive insights supporting human-systems integration and system-level decision-making.

Research Engineer - Firefighter-Robot Cooperation in Hostile Environments
LAMIH, Université Polytechnique Hauts-de-France - 2014–2015

- Identified key challenges during firefighter deployment in crisis scenarios through field observations and on-site interviews.
- Built a firefighter-robot teaming proof of concept to demonstrate potential for reducing risk in hostile environments.

Human Factors Researcher - Human-Centered Design (Aviation)
Florida Institute of Technology - 2014

- Prototyped and improved a digital flight information system for airline pilots using cognitive task analysis and human-factors assessments.
- Performed human-in-the-loop simulations and observational studies to evaluate pilot interaction in complex operational settings.
- Delivered findings supporting usability improvements and human-systems integration.

Human Factors Researcher
Eberspaecher - Saint-Michel, France - 2012

- Performed physical and cognitive ergonomics assessments to evaluate workstation design, operator posture, and task demands in an industrial environment.
- Applied ergonomic principles and task analysis to identify human-system mismatches and recommend design modifications improving operator comfort, safety, and performance.
- Supported prototyping and system documentation, including GRAFCET development, to enhance clarity of operational processes.

CORE COMPETENCIES

Human-Systems Integration & Human Factors & Ergonomics: Task/activity analysis · Cognitive engineering · Workload (NASA-TLX), Situation Awareness (SART) and Decision-Making assessment · Human-machine teaming

Human-Systems Interaction & Simulation: Scenario-based design · Prototyping · Human-in-the-loop simulations · Cockpit simulation · Test and evaluation planning

Qualitative Research Methods & Analysis: In-depth, semi-structured interviews · Grounded theory study · Usability testing · Heuristics evaluation · Pattern recognition and data synthesis

Prototyping & Tools: Functional cockpit prototyping during Ph.D. research (C++, Objective-C, Swift, MATLAB) · Figma · Miro · Jira & Confluence · Remote research platforms (UserTesting, Userlytics)

EDUCATION

Ph.D., Human-Centered Design (Aviation) - Florida Institute of Technology, 2018

Dissertation: Human-Centered Design of a Strategic 3D-Augmented Weather Management System for the Aviation Community (advisor: Dr. Lucas Stephane)

M.S., New Technologies, Transportation, and Security - Université Polytechnique Hauts-de-France, 2015

B.S., Engineering Science - Université Polytechnique Hauts-de-France, 2013

Publications: 8 publications, including conference papers, a journal article, and a book chapter, on cockpit weather decision support, onboard information systems, crew resource management, and design-time risk (IEA Congress, AHFE, HCI-Aero, Journal of Workplace Learning, CRC Press); full list on request

Languages: English (fluent), French (native)